Lower body specialization — Session A (Monday)

Hinge emphasis · hypermobility-safe · knee rehab phase · ~49 min · 22 working sets

Muscle group

Quads · Hamstrings · Glutes · Calves

30–60 min before: collagen peptides (10–15g) + vitamin C
Maximizes collagen synthesis in tendons loaded during session

B — Superset 1: Hinge + Quad iso (~9 min · 3 rounds · 90s rest after each pair)

1 DB Romanian deadlift (bilateral · soft knee bend) — 3 × 8–10
Hamstrings · glutes · erector spinae — DB · 3s ecc
Hip hinge · DBs slide along thighs · stop at mid-shin · primary hinge compound

↕ then immediately

2 Wall sit (isometric) — 3 × 30s
Quads · VMO (glutes/hams as iso stabilizers only) — Bodyweight
Start shallow · only deepen if pain-free · neutral antagonist for RDL

C — Superset 2: Hip extension + Calf (~8 min · 3 rounds · 90s rest after each pair)

3 Barbell hip thrust (bench-supported) — 3 × 8–10
Glute max · hamstrings — BB · 3s ecc
Stop short of full hip extension · drive through heels · squeeze at top

↕ then immediately

4 Standing calf raises (Smith · optional deficit) — 3 × 12–15
(3s eccentric · 1s pause at bottom stretch · soft knee) — Smith
Gastrocnemius (primary) · soleus synergist
Optional deficit: plate under balls of feet · heel drop ~3–5cm · 70–80% range

D — Superset 3: Knee flexion + Soleus (~6 min · 3 rounds · 60s rest after each pair)

5 Prone hamstring curl — 3 × 12
Biceps femoris · semimembranosus · semitendinosus — Band / light DB
Lying flat · light load · no knee snap at bottom · different joint action than RDL

↕ then immediately

6 Seated calf raise (plate on knees) — 3 × 12–15
(3s eccentric · 1s pause at bottom stretch · soft lockout) — Plate on knees
Soleus — knee-flexed position targets soleus over gastroc

E — Finisher: Hip stability + Adductor (~5 min · straight sets · 45s rest)

7 Side-lying banded hip abduction — 3 × 12/side
Glute medius · hip abductors — Band
Knee stabilizer maintenance · isolation finisher

8 Copenhagen plank holds (bench-supported) — 2 × 15s/side
(top leg on bench · bottom leg suspended · 20–30% effort) — Bodyweight
Adductors · isometric in lengthened position

Within 60 min after: protein 30–40g + vitamin C
Supports muscle + tendon remodeling · aim 1.6–2g protein/kg/day total

Every rep, every set — hypermobility + knee rehab rules

- Never lock out knees — soft bend at end range
- No single-leg eccentric loading through knee flexion-extension cycles (rehab phase)
- Skip session if joints feel unusually loose that day — reschedule, don't push
- Stop set early if any joint feels unstable — prioritize joint feel over rep count
- No pain rule — stop immediately if any knee pain or swelling
- Lengthened-position work targets ~70–80% of available range — never end range
- Hip abductor/adductor machine: deferred until knee stabilizes

Session timeline

Superset B (9 min) → Superset C (8 min) → Superset D (6 min) → Finisher E (5 min)
Total working session: ~49 min · 22 working sets · then proceed to post-workout mobility protocol

Working set count — Session A

Hamstrings: 6 sets (RDL 3 + curl 3)
Glutes: 3 direct (hip thrust) + RDL synergy
Quads: 3 sets (wall sit iso)
Calves: 6 sets (standing 3 + seated 3)
Hip abductors: 3 sets (side-lying)
Adductors: 2 sets (Copenhagen iso)

Alternative exercises — if Smith machine occupied

# Primary exercise	Alternative 1	Alternative 2
4 Standing calf raises (Smith)	DB standing calf raise (hold rack)	Single-leg DB calf raise (hold rack)

v3 — warmup removed (redundant with mobility doc) · alts audit (BB calf raise → single-leg DB)